

Amadreza Farahani | ML/AI Engineer

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Skills

Agentic AI	LangGraph, LangChain, LangSmith, MCP, multi-agent, agentic RAG (hybrid retrieval, reranking), guardrails
LLM Systems	Transformers, HuggingFace, AWS Bedrock, Ollama, vLLM, TorchServe, Milvus
Machine Learning	NumPy, Pandas, scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, MLflow, GANs, U-Net, YOLO, ASR
Cloud Technologies	AWS (Bedrock, ECS, EKS, S3, SageMaker), Docker
Software Development	Python, C/C++, SQL, Bash, Linux, Git, CI/CD, FastAPI, Prometheus, Grafana, Airflow

Professional Experience

AI Engineer

Zirak

Turin

11/2024 – Current

- Developed a production NL-to-SQL agent for **Poste Italiane**'s logistics division using **LangGraph** with **MCP**-exposed database tools, routing tool calls across **PostgreSQL**, MySQL, and MongoDB; FastAPI with **Ollama** for on-prem inference, cutting query authoring time by ~70% for non-technical analysts.
- Led a team of two to ship a cross-platform multi-agent meeting assistant (Electron/React + Python/FastAPI) combining real-time transcription, diarization, speaker verification, and **agentic RAG** with hybrid retrieval and reranking (**LangChain**/LangGraph) on **AWS Bedrock**, **LangSmith** tracing and regression evals, reducing summarization hallucinations by ~25%.
- Deployed an LLM assistant for consultancy and HR workflows, serving **HuggingFace** models via **vLLM**/**TorchServe** behind **FastAPI** with **Bedrock Guardrails** and **LangSmith** for traceability; saved ~50 hours/month of manual document review, enabling the product for additional clients.
- Designed a custom U-Net for multispectral imagery with an **Airflow**-orchestrated ingestion, training and evaluation pipeline on **AWS**, outperforming Random Forest, **XGBoost** and **LightGBM** baselines by ~10% F1 and reducing manual image review by ~80%.
- Built a rail-safety Computer Vision pipeline (**YOLOv8**/**PyTorch**) to detect sign defects and segment/track rails from nadir and frontal views in production.

Visiting Researcher – Master Thesis

Technische Universität Darmstadt | CROSSING

Darmstadt

08/2023 – 11/2024

- Designed a Diffusion **Transformer** for voice conversion (Python, PyTorch) on **HPC** GPUs, improving speaker similarity by ~9% over open-source baselines with fewer sampling steps, enabling real-time conversion.

Associate Generative AI Engineer

AROL S.p.a

Canelli

09/2022 – 09/2023

- Developed a synthetic machinery data generator (TensorFlow + **Flask** + **MongoDB**) and a customer testing dashboard, packaged with **Docker Compose** (client/server/DB) and REST API docs to avoid exposing proprietary data.
- Designed KPIs and Python tests to validate synthetic vs. real machinery signals and API responses, integrating these checks into the team's **CI pipeline** and reducing manual QA effort for customer demos.
- Reviewed over 5,000 lines of production **Python** and ML pipeline code (training, inference, data preprocessing), adding tests and reducing compute cost while making models easier to debug and extend.

Machine Learning Engineer

Part AI Research Center

Tehran

02/2020 – 03/2022

- Refined and fine-tuned an on-device keyword-spotting Android app in **Java**, using hard negative samples to reduce false positives by ~30%.
- Optimized a speaker verification system on **TensorFlow**, fine-tuned on new domains that reduced the Equal Error Rate (EER) by ~2%.
- Engineered **Test-Driven Development TDD** for Persian **ASR** and **NLP** services; wrote unit/integration tests to reach 100% coverage and set up **Jenkins CI/CD** to automate builds, tests, and releases.
- Mentored new team members on clean, production-ready **Python** (structure, testing, logging), improving code quality and safety.
- Implemented **Neural Network** architectures for scalable products, optimizing for both cost-efficiency and operational safety.

Internship Experience

Computer Vision - Bachelor Thesis

Megamouj Co. | University of Science and Technology (IUST)

Tehran

09/2019 – 02/2020

- Automated a traffic-flow monitoring system (**C++**/Python, **OpenCV**, YOLOv3) for vehicle detection, tracking, and flow estimation, reducing manual video review in a national-scale project.
- Engineered a data analysis pipeline using **NumPy** and **Pandas** to structure annotation datasets, utilizing **scikit-learn** to compute performance metrics and validate system reliability.
- Labeled and augmented traffic video data and trained/evaluated detection and tracking models in **Python**, tuning YOLOv3 hyperparameters and metrics to keep counting errors within acceptable limits across varying lighting and weather conditions.

Education

M.Sc. Computer Engineering - Artificial Intelligence & Data Analytics

Politecnico di Torino

Turin

2022 - 2024

- Thesis:** RobVC: Robust Zero-Shot Self Supervised Voice Conversion (Funded by CROSSING Co. in Darmstadt) | [Blog](#)

B.Sc. Electrical and Computer Engineering - Telecommunications

University of Science and Technology (IUST)

Tehran

2016 – 2021

- Thesis:** Deep Learning-Based Vehicle Detection and Traffic Flow Analysis (Funded by Megamouj Co.) | [Blog](#)